

YUHESEN Commercial Navigation API Interface Description

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Interface Version: v1.4.4

1. Project Overview

Introduction: This document describes the HTTP interface interaction specification for YUHESEN Magic commercial navigation robots, used for communication between host computers or third-party systems and the robot navigation system.

2. Service Environment Description

Environment Name	Base URL (Interface Prefix)	Description
Default Environment	http://192.168.1.102:8888/yhs-robot/	Default service address inside the robot

3. Global Specifications

3.1 Authentication

- Method: Bearer Token / Basic Auth / None

3.2 General Response Format

All system interfaces adopt a unified JSON response format:

```
{
  "errCode": 0,           // 错误码, 0 表示成功, 非 0 表示异常
  "msg": "success",       // 接口响应信息或错误提示
  "succeeded": true,      // 请求是否处理成功
  "data": { ... }         // 具体的业务数据载荷
}
```

4. Interface Details

4.1 Login Authentication

4.1.1 Login to Get Token

Method: POST

URL: /auth/token

Body:

```
{
  "userCode": "admin",
  "password": "admin123"
}
```

Response Example:

```
{
  "data": {
    "expires": "2024-12-01 14:21:06",
    "mac_address": [ // mac 地址
      "c4:83:72:10:cf:9c",
      "c4:83:72:10:cf:9d"
    ],
    "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJqdGkiOiI2MjUyNzY0NC04ZmU3LTRlYmMtYTZjYi1hYmRmZGI4YTA2OGUiLCJleHAiOjE3MzMwMzQwNjYsImhhdCI6MTczMDQ0MjA2NiwidHlwZSI6InJlZnJlc2giLCJuYmYiOjE3MzA0NDIwNjYsImlkZW50aXR5IjoieWRtaW4ifQ.jomMXe1-EidKmhngzI2aGqkrkUUy0MSVzbwmvP2q4-k", // 未使用
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJqdGkiOiI1ODYxOWJhMC04Y2U1LTRhZjYtODg4Mi0wYjM0YzExMWRhZTk1LCJleHAiOjE3MzMwMzQwNjYsImZyZXNoIjpmYWxzZSwiaWF0IjoxNzMwNDQyMDY2LCJ0eXB1IjoieWNjZXNzIiwibmJmIjoxNzMwNDQyMDY2LCJpZGVudG10eSI6eyJ1c2VyTmFtZSI6ImFkbWluIiwidXNlcklkIjo2LCJyb2xlc2VudG10eSI6eyJ1c2VyTmFtZSI6ImFkbWluIn19.UlhdtSfVMebvqB9tCHhbz-QK9NjQtS0x-jbX7Bv90Qo", // token 内容
    "userCode": "admin",
    "userId": 6,
    "userName": "admin",
    "version": "1.4.0" // 后端版本
  },
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

4.1.2 Refresh Token (unused)

4.2 Map Module

4.2.1 Start Mapping

Method: POST

URL: /cmd/begin_map?map_name={map_name}

Query:

Parameter	Required	Type	Description
-----------	----------	------	-------------

Parameter	Required	Type	Description
map_name	Yes	String	Map name

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "开启新建地图节点成功",
  "succeeded": true
}
```

Failure Example:

```
{
  "data": null,
  "errCode": 401,
  "msg": "token 已过期",
  "succeeded": false
}
```

Note:

- After calling the mapping interface, the program will take some time to start. Wait for the interface response or until the red laser point cloud appears on the APP, then use the remote control to move the car for mapping.

4.2.2 Get Real-time Mapping Image PNG (Called during mapping)

Method: GET

URL: /real_time_data/file_map

Response Example:

```
{
  "data": {
    "py/b64": "py/b64": "iVBORw0KGgoAAAANSUhEUgAABAMAAAHUCAIAAADA33ZJAAAgAE1EQVR4AezBW25qTa9A0bkk98H15Op/s8pPdhuwxNlan/gFJyGB3BM8xrbiQGuttdZaa+3BbCs0tNZaa6211h7MtuJAa6211lpr7cFsKw601lprrbXWHsy24kBrrbXWWmvtwWwrDrTWmuttdYezLbiQGuttdZaa+3BbCs0tNZuM4ewW1G01lprrf1m24oDrbX7zSGcrChaa6211n6VbcWB1lprrbXW2oPZVhXorbXWWmutPZhtxYHWWmuttdb....."
  },
  "errCode": 0,
  "msg": "数据获取成功",
  "succeeded": true
}
```

Note:

- The content in “data” is Base64 format data, which needs to be converted to an image format for display.

4.2.3 Stop Scanning and Save Map (Called during mapping) (Asynchronous)

Method: GET

URL: /cmd/async_stop_scan_map

Response Example:

```
{
  "data": {
    "createdAt": "2025/11/08 10:09:38",
    "gridHeight": 723, // 地图栅格高度
    "gridWidth": 621, // 地图栅格宽度
    "grid_x": 151.49836505206878, // 地图世界坐标系 (world) 原点在栅格坐标系下的 X 坐标 (只取整数部分)
    "grid_y": 355.6333870346339, // 地图世界坐标系 (world) 原点在栅格坐标系下的 Y 坐标 (只取整数部分)
    "originX": -7.574918365478516, // 地图栅格坐标系 (grid) 原点在世界坐标系下的 X 坐标
    "originY": -17.78166961669922, // 地图栅格坐标系 (grid) 原点在世界坐标系下的 Y 坐标
    "resolution": 0.05000000074505806 // 地图分辨率
  },
  "errCode": 0,
  "msg": "保存地图成功",
  "succeeded": true
}
```

Failure Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "保存地图成功,设置默认地图-错误:cannot concatenate 'str' and 'NoneType' objects",
  "succeeded": true
}
```

Note:

- **Caution:** Do not call this interface when no mapping is in progress, as it will cause a bug, as shown in the failure example above; the APP will display: subscribe node start Failed.

- For a map, the starting point of the mapping is the world coordinate origin, the front of the car is the X-axis, following the right-hand coordinate system, and the lower-left corner of the map is the grid coordinate origin.
- After calling this interface to stop scanning the map, the navigation program will automatically restart. During this process, do not use the remote control to move the car; it is recommended to wait a moment before proceeding.

4.2.4 Cancel Scanning Without Saving Map (Called during mapping)

Method: GET

URL: /cmd/cancel_scan_map

Response Example:

```
{
  "data": true,
  "errCode": 0,
  "msg": "取消成功",
  "succeeded": true
}
```

Note:

- After calling this interface to cancel scanning the map, the navigation program will automatically restart and use the previous map. During this process, do not use the remote control to move the car; it is recommended to wait a moment before proceeding.

4.2.5 Get Map Image in PNG Format

Method: GET

URL: /data/get_map_png?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name

Response Example:

```
{
  "data": "iVBORw0KGgoAAAANSUhEUgAAAcwAAAP4CAIAAADfzVtDAAAgAE1EQVR4AezBW24b2w4FwEWAc2B/sec/LPKLHMMm4Bso8IF9HSd+SC1LWlUstUBERJchUQtERHQZErVAR  
ESXIVELRER0GRK1QERELyFRC0REdBkStUBERJchUQtERHQZErVA927fNGpARIEtqAw60/um  
UbNvCiBq9k2jBkR0OI1aoDuyb4qTqAERXZtELdAN2jeNmn3TqAER/VQStUC3Y98UQNTsm0b  
NvmnUgIh+KolaoFuwbxo1+J5906g.....",
  "errCode": 0,
}
```

```
"msg": "数据获取成功",  
"succeeded": true  
}
```

Note:

- This interface is used to get the complete map image.

4.2.6 Rename Map (Will restart navigation service)

Method: GET

URL:

/cmd/rename_map?origin_map_name={origin_map_name}&new_map_name={new_map_name}

Query:

Parameter	Required	Type	Description
origin_map_name	Yes	String	Original map name
new_map_name	Yes	String	New map name

Response Example:

```
{  
  "data": true,  
  "errCode": 0,  
  "msg": "修改成功",  
  "succeeded": true  
}
```

Note:

- Note that after renaming the currently used map, the navigation program will automatically restart. Please wait a moment, and do not use the remote control to move the car during this time.

4.2.7 Get Map List

Method: GET

URL: /data/list_maps

Response Example:

```
{  
  "data": [  
    {  
      "createdAt": "2024/11/11 11:02:11",  
    }  
  ]  
}
```

```
    "creator": "董小菰",
    "default": true, // 是否是当前正在使用的地图
    "id": "573be798-9fd9-11ef-b5c3-c4837210cf9c", // 地图 id, 对应 yhs-robot/datas 地图数据里的文件夹名称
    "mapInfo": {
      "createdAt": "2024/11/11 11:02:11",
      "gridHeight": 734, // 地图栅格高度
      "gridWidth": 600, // 地图栅格宽度
      "originX": -8.35, // 栅格原点 X 坐标
      "originY": -17.38670654296875, // 栅格原点 Y 坐标
      "resolution": 0.05000000074505806 // 地图分辨率
    },
    "name": "map" // 地图名称
  },
  ],
  "errCode": 0,
  "msg": "数据获取成功",
  "succeeded": true
}
```

4.2.8 Show Specified Map Information

Method: GET

URL: /data/single_map_info?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Specified map name

Response Example:

```
{
  "data": {
    "createdAt": "2024/11/11 11:02:11",
    "gridHeight": 734,
    "gridWidth": 600,
    "mapname": "in",
    "mapsize": "2.9M", // 地图大小
    "originX": -8.35,
    "originY": -17.38670654296875,
    "resolution": 0.05000000074505806
  },
  "errCode": 0,
  "msg": "数据获取成功",
}
```

```
"succeeded": true
}
```

4.2.9 Get Currently Used Default Map

Method: GET

URL: /cmd/use_map

Response Example:

```
{
  "data": {
    "createdAt": "2024/11/11 11:02:11", // 地图创建时间
    "creator": "董小菰", // 创建者
    "default": true, // 是否为默认地图
    "id": "573be798-9fd9-11ef-b5c3-c4837210cf9c", // 地图 ID
    "mapInfo": {
      "createdAt": "2024/11/11 11:02:11",
      "gridHeight": 734, // 地图栅格高度和宽度
      "gridWidth": 600,
      "originX": -8.35, // 原始坐标值,在地图左下角
      "originY": -17.38670654296875,
      "resolution": 0.05000000074505806 // 地图分辨率
    },
    "name": "in" // 地图名字
  },
  "errCode": 0,
  "msg": "",
  "succeeded": true
}
```

4.2.10 Delete Map

Method: GET

URL: /cmd/delete_map?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Name of the map to delete

Response Example:

```
{
  "data": null,
  "errCode": 0,
}
```



```
"msg": "地图删除成功",
"successed": true
}
```

Failure Example:

```
{
  "data": null,
  "errCode": -1,
  "msg": "地图正在使用中!",
  "successed": false
}
```

4.2.11 Switch Map

Method: POST

URL: /cmd/use_map?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Name of the map to switch to

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "开启 navigation.launch0619-4 成功",
  "successed": true
}
```

Note:

- Note that after switching the map, the navigation program will automatically restart. Please wait a moment, and do not use the remote control to move the car during this time.

4.2.12 Draw Special Areas

Method: POST

URL: /obstacle?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Name of the map on which

Parameter	Required	Type	Description
			to draw areas

Body:

```
{
  "carpets": { // 地毯区域(未使用)
    "circles": [],
    "lines": [],
    "polygons": [],
    "polylines": [],
    "rectangles": []
  },
  "decelerations": { // 减速带区域
    "circles": [],
    "lines": [],
    "polygons": [ // 多边形减速带
      [
        {
          "x": 527,
          "y": 366
        },
        {
          "x": 538,
          "y": 366
        },
        {
          "x": 537,
          "y": 401
        },
        {
          "x": 524,
          "y": 400
        }
      ]
    ],
    "polylines": [],
    "rectangles": []
  },
  "displays": { // 窄道
    "circles": [],
    "lines": [],
    "polygons": [ // 多边形窄道
      [
        {
          "x": 617,
          "y": 322
        }
      ]
    ]
  }
}
```

```

        "x": 620,
        "y": 277
    },
    {
        "x": 630,
        "y": 277
    },
    {
        "x": 630,
        "y": 312
    }
]
],
"polylines": [],
"rectangles": []
},
"highlight": { // 未使用
    "circles": [],
    "lines": [],
    "polygons": [],
    "polylines": [],
    "rectangles": []
},
"obstacles": { // 虚拟墙
    "circles": [ // 圆形虚拟墙
        {
            "center": { // 圆心栅格坐标
                "x": 663,
                "y": 251
            },
            "radius": 13 // 栅格半径
        }
    ],
    "lines": [],
    "polygons": [ // 多边形虚拟墙
        [
            {
                "x": 637, // 多边形顶点栅格坐标
                "y": 378
            },
            {
                "x": 680,
                "y": 375
            },
            {
                "x": 679,
                "y": 361
            },
            {

```

```

        "x": 636,
        "y": 362
    }
    ],
    "polylines": [ // 线段虚拟墙
        [
            {
                "x": 634,
                "y": 444
            },
            {
                "x": 666,
                "y": 444
            }
        ]
    ],
    "rectangles": []
},
"slopes": { // 斜坡区域
    "circles": [],
    "lines": [],
    "polygons": [ // 多边形斜坡
        [
            {
                "x": 637, // 多边形顶点栅格坐标
                "y": 355
            },
            {
                "x": 661,
                "y": 351
            },
            {
                "x": 665,
                "y": 367
            },
            {
                "x": 642,
                "y": 369
            }
        ]
    ]
}
}

```

Response Example:

```

{
    "data": "保存成功",
}

```

```
"errCode": 0,
"msg": "success",
"successed": true
}
```

Note:

- Note: Do not use fields in the Body example above that do not have actual coordinate data filled in. For example, only the “polygons” field is used in slopes.
- After drawing and saving areas, they take effect immediately when a navigation task is sent. During navigation, the corresponding effect occurs only when the robot is inside the area.
- After clearing virtual walls, it takes effect immediately without restart. After deleting slope areas, deceleration zones, and narrow passages, a restart of the navigation system is required for it to take effect.
- All coordinate data in the fields use grid values.

4.2.13 Get Special Area Data

Method: GET

URL: /obstacle?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name

Response Example:

```
{
  "data": {
    "carpets": {
      "circles": [
        {}
      ],
      "lines": [
        {}
      ],
      "polygons": [
        {}
      ],
      "polylines": [
        {}
      ],
      "rectangles": [
        {}
      ]
    }
  }
}
```

```
    },
    "decelerations": {
      "circles": [
        {}
      ],
      "lines": [
        {}
      ],
      "polygons": [
        {}
      ],
      "polylines": [
        {}
      ],
      "rectangles": [
        {}
      ]
    },
    "displays": {
      "circles": [
        {}
      ],
      "lines": [
        {}
      ],
      "polygons": [
        {}
      ],
      "polylines": [
        {}
      ],
      "rectangles": [
        {}
      ]
    },
    "highlight": {
      "circles": [
        {}
      ],
      "lines": [
        {}
      ],
      "polygons": [
        {}
      ],
      "polylines": [
        {}
      ],
      "rectangles": [
        {}
      ]
    }
  }
}
```

```
    ],
    "obstacles": {
      "circles": [
        {
          "center": {
            "x": 34,
            "y": 45
          },
          "radius": 88
        }
      ],
      "lines": [
        {
          "end": {
            "x": 78,
            "y": 30
          },
          "start": {
            "x": 10,
            "y": 66
          }
        }
      ],
      "polygons": [
        [
          {
            "x": 41,
            "y": 87
          }
        ]
      ],
      "polylines": [
        [
          {
            "x": 27,
            "y": 7
          }
        ]
      ],
      "rectangles": [
        {
          "end": {
            "x": 50,
            "y": 5
          },
          "start": {
            "x": 72,
            "y": 43
          }
        }
      ]
    }
  },
  "obstacles": {
    "circles": [
      {
        "center": {
          "x": 34,
          "y": 45
        },
        "radius": 88
      }
    ],
    "lines": [
      {
        "end": {
          "x": 78,
          "y": 30
        },
        "start": {
          "x": 10,
          "y": 66
        }
      }
    ],
    "polygons": [
      [
        {
          "x": 41,
          "y": 87
        }
      ]
    ],
    "polylines": [
      [
        {
          "x": 27,
          "y": 7
        }
      ]
    ],
    "rectangles": [
      {
        "end": {
          "x": 50,
          "y": 5
        },
        "start": {
          "x": 72,
          "y": 43
        }
      }
    ]
  }
}
```

```

    }
  ],
  "slopes": {
    "circles": [
      {}
    ],
    "lines": [
      {}
    ],
    "polygons": [
      [
        {
          "x": 5,
          "y": 89
        }
      ]
    ],
    "polylines": [
      {}
    ],
    "rectangles": [
      {}
    ]
  }
},
"errCode": 0,
"msg": "success",
"successed": true
}

```

4.2.14 Original Map Editing (Will restart navigation service)

Method: POST

URL: /cmd/edit_map?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name

Body:

```

[
  // {
  //   "type": "polygons", // 多边形
  //   "operation_type": "add", // 操作类型 add: 填充 remove: 清除
  //   "datas": [
  //     {

```



```

//         "x": 705,
//         "y": 495
//     },
//     {
//         "x": 715,
//         "y": 495
//     },
//     {
//         "x": 705,
//         "y": 485
//     },
//     {
//         "x": 715,
//         "y": 485
//     }
// ]
// }
{
    "datas": {
        "center_x": 705,
        "center_y": 495,
        "radius": 15
    },
    "operation_type": "add",
    "type": "circles"
}
// {
//     "type": "polygons",
//     "operation_type": "remove",
//     "datas": [
//         {
//             "x": 341,
//             "y": 42
//         },
//         {
//             "x": 406,
//             "y": 49
//         },
//         {
//             "x": 399,
//             "y": 119
//         }
//     ]
// },
// {
//     "type": "circles",
//     "operation_type": "remove",
//     "datas": [
//         {
//             "radius": 42,

```

```
//          "center_x": 496,
//          "center_y": 74
//      }
//  ]
// }
```

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "操作成功",
  "succeeded": true
}
```

Note:

- Calling this interface will cause the navigation system to restart.
- Operations performed via this interface cannot be undone.

4.2.15 Dynamic Pose Initialization (Car rotates in place)

Method: POST

URL:

/real_time_data/dynamic_init_pose?map_name={map_name}&point_name={point_name}

Parameter	Required	Type	Description
map_name	Yes	String	Map name
point_name	Yes	String	Initial point name

Query:

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "正在动态初始化...",
  "succeeded": true
}
```

Note:

- Drive the car to the pre-created initialization point using the remote control, ensuring the car's heading matches the direction of the initialization point.

Then switch the remote control to navigation mode and call dynamic pose initialization. Indoor initialization positioning is very fast; outdoor initialization positioning may take slightly longer depending on environmental references.

(If you observe on the APP that the red laser points are aligned or slightly deviated from the surrounding obstacle edges, you can manually rotate the car in place using the remote control to quickly initialize successfully.)

- After calling this interface, differential-drive chassis will rotate in place for pose initialization; Ackermann chassis will move along an arc.
- During initialization, there must be no obstacles near the car, otherwise the car will not move.
- If the car's localization is accurate, initialization is not required.
- If the car's localization shows success but the actual localization is incorrect (mis-localization), re-initialization is needed.

4.2.16 Initialize Using Map Grid Coordinates (Car rotates in place)

Method: POST

URL: /real_time_data/temp_point_init_pose

Body:

```
{
  "angle": 23.55, // 机器人朝向角度
  "gridX": 503, // 栅格坐标
  "gridY": 304,
  "mapName": "in", // 地图名称
  "type": 1 // 0 静态初始化（未使用），1 动态初始化
}
```

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "正在使用地图坐标初始化....",
  "succeeded": true
}
```

4.2.17 Static Pose Initialization (unused)

Method: POST

URL:

/real_time_data/static_init_pose?map_name={map_name}&point_name={point_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name
point_name	Yes	String	Initial point name

4.2.18 Initialization Result

Method: GET

URL: /real_time_data/init_pose_result

Response Example:

```
{
  "data": true,
  "errCode": 0,
  "msg": "初始化成功...",
  "succeeded": true
}
```

Failure Example:

```
{
  "data": null,
  "errCode": -1,
  "msg": "初始化失败...",
  "succeeded": false
}
```

4.2.19 Recover Map (Under development)

Method: GET

URL: /cmd/recover_map?map_name={map_name}

4.3 Path and Point Management Module

4.3.1 Create Navigation Point

Method: POST

URL: /cmd/pos

Body:

```
{
  "angle": -0.0036086901577556674, // 导航点朝向角度 (-180°~180°)
  "gridX": 896, // 导航点栅格坐标
  "gridY": 489,
  "mapName": "q", // 地图名称
  "name": "a", // 导航点名称
  "type": 2, // 导航点类型, 目前只用到了 0, 1, 2
  // 初始化点: 0 可以用于初始化和导航
  // 充电点: 1 用于自主回充
  // 导航点: 2 用于导航
  // RFID 点: 3
  // 注水点: 4
  // 排水点: 5
  // 二维码点: 11
  // 导航结束点: 12
  "markPoint": false, // 是否标记当前位姿 (将小车当前的位姿作为导航点的位姿)
  "moveType": 1 // 导航模式 1: 前进导航, 2: 后退导航, 3: 横移导航
}
```

Response Example:

```
{
  "data": {
    "angle": 0, // 导航点角度 (°)
    "gridX": 384, // 导航点栅格坐标
    "gridY": 385,
    "id": "770c0c68-9757-11ef-9db5-c4837210cf9c", // 导航点 ID
    "mapId": "8b0d382e-9755-11ef-9db5-c4837210cf9c", // 地图 ID
    "mapName": "map", // 地图名称
    "markPoint": false, // 是否为标记当前点
    "moveType": 1, // 导航模式: 1 前进导航, 2 后退导航, 3 横移模式
    "name": "1", // 导航点名称
    "type": 2, // 导航点类型
    "worldPose": { // 导航点世界坐标
      "orientation": {
        "w": {
          "py/newargs": [
            1
          ]
        },
        "x": {
          "py/newargs": [
            0
          ]
        },
        "y": {
```

```

        "py/newargs": [
            0
        ],
        "z": {
            "py/newargs": [
                0
            ]
        },
        "position": {
            "x": 12.246883000000004,
            "y": 1.6918930000000003,
            "z": 0
        }
    },
    "errCode": 0,
    "msg": "添加成功",
    "succeeded": true
}

```

Note:

- When using the markPoint field to mark the current position, ensure the car's localization is accurate, otherwise the created point coordinates will be incorrect.
- Regarding navigation mode moveType:
 1. Forward navigation means the car moves towards the target point with the front facing the target.
 2. Backward navigation means the car moves towards the target point with the rear facing the target.
 3. Lateral navigation is a specific motion mode for four-wheel-steering chassis, meaning the car moves towards the target point in lateral control mode.

Forward and backward navigation modes are available for all vehicle models; lateral navigation mode is only available for four-wheel-steering chassis.

- When the markPoint field is true, the gridX, gridY, and angle fields must also match the car's current pose, otherwise the point displayed on the frontend APP will be incorrect.

4.3.2 Edit Navigation Point

Method: PUT

URL: /cmd/pos

Body:

```
{
  "angle": 112, // 修改后点的角度
  "gridX": 120, // 修改后点的栅格坐标
  "gridY": 1339,
  "mapName": "ab",
  "id": "91c662c8-2c72-11f0-93e0-c4837210cf9c", // 导航点 ID
  "name": "g", // 修改后的点的名称
  "markPoint": false, // 标记当前位置
  "type": 1, // 修改后的点的类型
  "moveType": 1 // 修改后点的导航模式 1: 前进导航, 2: 后退导航, 3: 横移导航
}
```

Response Example:

```
{
  "data": {
    "angle": 112,
    "gridX": 120,
    "gridY": 1339,
    "id": "770c0c68-9757-11ef-9db5-c4837210cf9c",
    "mapId": "8b0d382e-9755-11ef-9db5-c4837210cf9c",
    "mapName": "map",
    "markPoint": false,
    "name": "p1",
    "type": 0,
    "worldPose": {
      "orientation": {
        "w": {
          "py/newargs": [
            0.5591929034707468
          ]
        },
        "x": {
          "py/newargs": [
            0
          ]
        },
        "y": {
          "py/newargs": [
            0
          ]
        },
        "z": {
          "py/newargs": [
            0.8290375725550417
          ]
        }
      }
    }
  }
}
```

```

    ],
    },
    "position": {
      "x": -0.9531169999999998,
      "y": 49.391893,
      "z": 0
    }
  },
  "errCode": 0,
  "msg": "编辑成功",
  "succeeded": true
}

```

Note:

- Note: When using this interface, if some field attributes are not filled in, they will automatically become default attributes. For example, if the original moveType was 2, and moveType is commented out during editing of the navigation point, it will default to 1.

4.3.3 Delete Navigation Point

Method: DELETE

URL: /cmd/pos?map_name={map_name}&position_name={point_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name
position_name	Yes	String	Point name

Response Example:

```

{
  "data": null,
  "errCode": 0,
  "msg": "删除成功",
  "succeeded": true
}

```

4.3.4 Get Navigation Point List on Map

Method: GET

URL: /data/poslist?map_name={map_name}&type=

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name
type	No	Integer	Point type (returns all types when not passed)

Response Example:

```
{
  "data": [
    {
      "angle": 0,
      "createdAt": "2023-07-12 10:07:37",
      "gridX": 592,
      "gridY": 190,
      "id": 0,
      "mapId": "926a94e8-2057-11ee-9650-042b58098dc5",
      "mapName": "0712",
      "name": "001",
      "type": 0,
      "worldPose": {
        "orientation": {
          "w": 1,
          "x": 0,
          "y": 0,
          "z": 0
        },
        "position": {
          "x": 15.514233,
          "y": -3.5,
          "z": 0
        }
      }
    }
  ],
  "errCode": 0,
  "msg": "获取成功",
  "succeeded": true
}
```

4.3.5 Generate Hand-Drawn Path

Method: POST

URL: /cmd/graph

Body:

```
{
  "name": "path1", // 手绘路径名称
```

```

"mapId": "435a63cc-ae41-11f0-9c9d-c4837210cf9d", // 地图 ID
"mapName": "t1", // 地图名称
"points": [
  {
    "actions": [
      {
        "fields": [
          {
            "fields": [],
            "name": "millisecond",
            "type": "3",
            "value": "3000" // 停留 3000ms
          }
        ],
        "isChecked": false,
        "isDevice": false,
        "name": "Pause"
      },
      {
        "fields": [
          {
            "fields": [],
            "name": "rotation",
            "type": "double",
            "value": "0" // 旋转到的角度° 范围：0°~360°
          }
        ],
        "isChecked": false,
        "isDevice": false,
        "name": "Rotate2d" // 旋转
      }
    ],
    "name": "1", // 手绘路径第一个点的名称
    "gridPosition": { // 手绘路径第一个点的栅格坐标
      "x": 191,
      "y": 386
    },
    "angle": 0, // 点的朝向 -180°~180°，这里没有使用，如果要使用角度，在
actions 里添加动作点
    "moveType": 1 // 导航模式 1: 前进模式 2: 后退模式 3: 横移模式
  },
  {
    "actions": [],
    "name": "2", // 手绘路径第二个点的名称
    "gridPosition": {
      "x": 191,
      "y": 466
    },
  },

```

```

        "angle": 0,
        "moveType": 1
    }
],
"lines": [
    {
        "radius": 0, // radius 为 0, 手绘路径为直线, 不为 0, 手绘路径为弧线, 曲
        // 率为正, 手绘路径朝前进方向的左边弯曲
        "begin": "1",
        "end": "2",
        "name": "1_2" // 手绘路径中线段的名称
    }
],
"pathGroups": [],
"paths": [
    {
        "lines": [
            {
                "name": "1_2", // 手绘路径线段的名称
                "reverse": false
            }
        ],
        "points": [
            {
                "actions": [],
                "name": "1",
                "gridPosition": {
                    "x": 191,
                    "y": 386
                },
                "angle": 0,
                "moveType": 1
            },
            {
                "name": "2",
                "actions": [],
                "gridPosition": {
                    "x": 191,
                    "y": 466
                },
                "angle": 0,
                "moveType": 1
            }
        ],
        "name": "path1_path0"
    }
],
"type": 3,

```

```
"close": false // 是否为闭环路径（至少三个点才能形成闭环路径）
}
```

Response Example:

```
{
  "data": {
    "close": false, // 是否是闭环路径
    "fileName": "47481fff-9765-11ef-9db5-c4837210cf9c.json", // 后端 yhs-robot 文件夹中路径 json
    "id": "47481ffe-9765-11ef-9db5-c4837210cf9c", // 手绘路径 id
    "length": 1.5, // 手绘路径线段长度
    "lines": [
      {
        "begin": "p1",
        "end": "p2",
        "length": 1.5,
        "name": "p1-p2",
        "radius": 0
      }
    ],
    "mapId": "8b0d382e-9755-11ef-9db5-c4837210cf9c",
    "mapName": "mapa",
    "name": "path",
    "pathGroups": [],
    "paths": [
      {
        "lines": [
          {
            "name": "p1-p2",
            "reverse": false
          }
        ],
        "name": "path1",
        "points": [
          {
            "actions": [
              {
                "fields": [
                  {
                    "fields": [],
                    "name": "millisecond",
                    "type": "3",
                    "value": "3000" // 停留时间(单位 ms)
                  }
                ]
              }
            ]
          }
        ],
        "isChecked": false,
        "isDevice": false,
      }
    ]
  }
}
```

```

        "name": "Pause"
    },
    {
        "fields": [
            {
                "fields": [],
                "name": "rotation",
                "type": "double",
                "value": "0" // 要旋转到的角度
            }
        ],
        "isChecked": false,
        "isDevice": false,
        "name": "Rotate2d"
    }
],
"angle": 0,
"gridPosition": {
    "x": 1397,
    "y": 2247
},
"moveType": 1,
"name": "p1",
"worldPose": {
    "orientation": {
        "w": 1,
        "x": 0,
        "y": 0,
        "z": 0
    },
    "position": {
        "x": 62.896883000000001,
        "y": 94.791893000000002,
        "z": 0
    }
}
},
{
    "actions": [
        {
            "fields": [
                {
                    "fields": [],
                    "name": "millisecond",
                    "type": "3",
                    "value": "3000"
                }
            ]
        }
    ]
}

```

(范围 $0^{\circ}\sim 360^{\circ}$)

```

        "isChecked": false,
        "isDevice": false,
        "name": "Pause"
    },
    {
        "fields": [
            {
                "fields": [],
                "name": "rotation",
                "type": "double",
                "value": "0"
            }
        ],
        "isChecked": false,
        "isDevice": false,
        "name": "Rotate2d"
    }
],
"angle": 0,
"gridPosition": {
    "x": 1367,
    "y": 2247
},
"moveType": 2,
"name": "p2",
"worldPose": {
    "orientation": {
        "w": 1,
        "x": 0,
        "y": 0,
        "z": 0
    },
    "position": {
        "x": 61.39688300000001,
        "y": 94.79189300000002,
        "z": 0
    }
}
}
}
}
],
"points": [
    {
        "actions": [
            {
                "fields": [
                    {
                        "fields": [],
                        "name": "millisecond",

```

```

        "type": "3",
        "value": "3000"
    }
    ],
    "isChecked": false,
    "isDevice": false,
    "name": "Pause"
},
{
    "fields": [
        {
            "fields": [],
            "name": "rotation",
            "type": "double",
            "value": "0"
        }
    ],
    "isChecked": false,
    "isDevice": false,
    "name": "Rotate2d"
}
],
"angle": 0,
"gridPosition": {
    "x": 1397,
    "y": 2247
},
"moveType": 1,
"name": "p1"
},
{
    "actions": [
        {
            "fields": [
                {
                    "fields": [],
                    "name": "millisecond",
                    "type": "3",
                    "value": "3000"
                }
            ],
            "isChecked": false,
            "isDevice": false,
            "name": "Pause"
        },
        {
            "fields": [
                {
                    "fields": [],
                    "name": "rotation",

```

```

        "type": "double",
        "value": "0"
    },
    ],
    "isChecked": false,
    "isDevice": false,
    "name": "Rotate2d"
},
],
"angle": 0,
"gridPosition": {
    "x": 1367,
    "y": 2247
},
"moveType": 2,
"name": "p2"
},
],
"type": 3
},
"errCode": 0,
"msg": "生成成功",
"successed": true
}

```

Failure Example 1:

```

{
    "msg": "The browser (or proxy) sent a request that this server could not understand.",
    "data": "POST /yhs-robot/cmd/graph",
    "errCode": -1,
    "successed": false
} // Body 内容格式错误

```

Failure Example 2:

```

{
    "data": null,
    "errCode": -1,
    "msg": "createPoints-错误:too many indices for array",
    "successed": false
} // 线段曲率半径 radius 错误

```

Failure Example 3:

```

{
    "msg": "negative number cannot be raised to a fractional power",
    "data": "POST /yhs-robot/cmd/graph",
    "errCode": -1,

```



```
"succeeded": false
} // 数据内容不一致
```

Note:

- In the interface, the arc radius 'radius' can be solved for the minimum arc radius using the grid coordinates of points 1 and 2. The value of radius can only be greater than the minimum arc radius, not smaller; otherwise, an error will be reported: createPoints-错误:too many indices for array
- Note that in the interface, the data for the same point must be filled in consistently. If there are discrepancies, an error will be reported: negative number cannot be raised to a fractional power
- In the interface, if the hand-drawn path has only two points, but 'close' is still set to true, an error will be reported: negative number cannot be raised to a fractional power
- When generating a hand-drawn path, the minimum distance between two points on a line segment is not less than 10cm, i.e., two grids.

4.3.6 Update Hand-Drawn Path (Update path action points)

Method: PUT

URL: /cmd/graph

Body:

```
{// 先通过“获取手绘路径”接口，获取字段，填入该接口中，再修改路径动作点
  "close": false, // 是否为闭环路径
  "coverageArea": null,
  "createAt": "2024-08-21 11:20:58",
  "data": {},
  "fileName": "62a89b07-5f6c-11ef-82c9-c4837210cf9c.json", // 后端中，
  路径的数据文件
  "id": "62a89b06-5f6c-11ef-82c9-c4837210cf9c",
  "length": 4.004996878900157, // 路径长度
  "lines": [
    {
      "angle": -20,
      "begin": "1", // 线段起点
      "end": "2", // 线段终点
      "length": 2.0024984394500787, // 线段长度
      "name": "1_2", // 线段名称
      "radius": 0 // 线段曲率半径
    },
    {
      "angle": -20,
      "begin": "2",
      "end": "3",
```

```

        "length": 2.0024984394500787,
        "name": "2_3",
        "radius": -20.0024984394500787
    }
],
"mapId": "ce4c2ca0-5f5e-11ef-82c9-c4837210cf9c", // 地图 ID
"mapName": "in", // 地图名称
"name": "ad", // 手绘路径名称
"paths": [
    {
        "lines": [
            {
                "name": "1_2",
                "reverse": false
            },
            {
                "name": "2_3",
                "reverse": false
            }
        ],
        "name": "ad_path0", // 路径名称
        "points": [
            {
                "actions": [], // 动作
                "angle": 92.86240522611176, // 点的角度
                "gridPosition": { // 栅格坐标
                    "x": 424,
                    "y": 379
                },
                "name": "1",
                "worldPose": { // 世界坐标
                    "orientation": {
                        "w": 0.6892250659458445,
                        "x": 0,
                        "y": 0,
                        "z": 0.724547312790508
                    },
                    "position": {
                        "x": -0.12923899999999833,
                        "y": -0.12492200000000153,
                        "z": 0
                    }
                }
            },
            {
                "actions": [],
                "angle": 92.86240522611176,
                "gridPosition": {
                    "x": 422,

```

```

        "y": 419
    },
    "name": "2",
    "worldPose": {
        "orientation": {
            "w": 0.6892250659458445,
            "x": 0,
            "y": 0,
            "z": 0.724547312790508
        },
        "position": {
            "x": -0.22923899999999975,
            "y": 1.875078000000002,
            "z": 0
        }
    }
},
{
    "actions": [],
    "angle": 92.86240522611176,
    "gridPosition": {
        "x": 424,
        "y": 379
    },
    "name": "3",
    "worldPose": {
        "orientation": {
            "w": 0.6892250659458445,
            "x": 0,
            "y": 0,
            "z": 0.724547312790508
        },
        "position": {
            "x": -0.12923899999999833,
            "y": -0.12492200000000153,
            "z": 0
        }
    }
}
],
"pointCount": null,
"points": [
    {
        "actions": [],
        "angle": 92.86240522611176,
        "gridPosition": {
            "x": 424,
            "y": 379
        }
    }
]

```

```

        },
        "name": "1"
    },
    {
        "actions": [],
        "angle": 92.86240522611176,
        "gridPosition": {
            "x": 422,
            "y": 419
        },
        "name": "2"
    },
    {
        "actions": [],
        "angle": 92.86240522611176,
        "gridPosition": {
            "x": 424,
            "y": 379
        },
        "name": "3"
    }
],
"type": 3
}

```

Response Example:

```

{
    "data": null,
    "errCode": 0,
    "msg": "更新手画路径成功",
    "succesed": true
}

```

4.3.7 Delete Hand-Drawn Path

Method: DELETE

URL:

/cmd/graph?map_name={map_name}&graph_path_name={graph_path_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name
graph_path_name	Yes	String	Hand-drawn path name

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "删除手画路径成功",
  "succeeded": true
}
```

4.3.8 Hand-Drawn Path List

Method: GET

URL: data/graph_list?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name

Response Example:

```
{
  "data": [
    {
      "close": null,
      "coverageArea": null,
      "createAt": "2025-12-08 16:17:00",
      "fileName": "45609da7-d40e-11f0-bd8a-c4837210cf9c.json",
      "id": "45609da6-d40e-11f0-bd8a-c4837210cf9c",
      "length": 2.507987240796891,
      "lines": [
        {
          "begin": "1",
          "end": "2",
          "length": 2.507987240796891,
          "name": "1_2",
          "radius": 0
        }
      ],
      "mapId": "f225aa88-d3f8-11f0-bd8a-c4837210cf9c",
      "mapName": "map1",
      "name": "s",
      "paths": [
        {
          "lines": [
            {
              "name": "1_2",
              "reverse": false
            }
          ],
          "name": "s_path0",

```

```

    "points": [
      {
        "actions": [
          {
            "fields": [
              {
                "fields": [],
                "name": "rotation",
                "type": "double",
                "value": "0"
              }
            ],
            "isChecked": false,
            "isDevice": false,
            "name": "Rotate2d"
          }
        ],
        "angle": -66.5014343240479,
        "gridPosition": {
          "x": 342,
          "y": 478
        },
        "moveType": 1,
        "name": "1",
        "worldPose": {
          "orientation": {
            "w": 0.8362792928843957,
            "x": 0,
            "y": 0,
            "z": -0.548303697135789
          },
          "position": {
            "x": 2.8263900000000017,
            "y": 1.5244140000000037,
            "z": 0
          }
        }
      },
      {
        "actions": [],
        "angle": -66.5014343240479,
        "gridPosition": {
          "x": 362,
          "y": 432
        },
        "moveType": 1,
        "name": "2",
        "worldPose": {
          "orientation": {
            "w": 0.8362792928843957,

```

```

        "x": 0,
        "y": 0,
        "z": -0.548303697135789
    },
    "position": {
        "x": 3.8263900000000017,
        "y": -0.7755859999999997,
        "z": 0
    }
}
]
},
"pointCount": null,
"points": [
    {
        "actions": [],
        "angle": -66.5014343240479,
        "gridPosition": {
            "x": 342,
            "y": 478
        },
        "moveType": 1,
        "name": "1"
    },
    {
        "actions": [],
        "angle": -66.5014343240479,
        "gridPosition": {
            "x": 362,
            "y": 432
        },
        "moveType": 1,
        "name": "2"
    }
],
"type": 3
},
"errCode": 0,
"msg": "success",
"successed": true
}

```

4.3.9 Get Hand-Drawn Path

Method: GET

URL:

data/graph_path?map_name={map_name}&path_name={graph_path_name}&contains_details=

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name
path_name	Yes	String	Hand-drawn path name
contains_details	No	String	True or False (whether to include coordinates of all points in the path. Can be omitted if not needed, pass True if needed)

4.3.10 Rename Hand-Drawn Path

Method: GET

URL:

/cmd/update_graph_path_name?map_name={map_name}&origin_name={origin_name}&new_name={new_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name
origin_name	Yes	String	Old hand-drawn path name
new_name	Yes	String	New hand-drawn path name

4.3.11 Check If a Line Segment Can Be Generated Between Two Points

Method: POST

URL: /cmd/graph_check_point

Body:

```
{
  "begin": {
    "x": 306,
    "y": 565
  },
  "end": {
    "x": 317,
```



```

        "y": 574
    },
    "mapName": "mapa",
    "radius": 0 // 线段的曲率半径，可通过两点的地图坐标求解最小的圆弧半径，radius 只能比最小圆弧半径值大。0 为直线（单位为栅格）
}
// 需要将导航参数 is_check_obs_in_path 设置为 true

```

Response Example:

```

{
    "data": null,
    "errCode": 0,
    "msg": "路径可以通行",
    "succeeded": true
}

```

Failure Example:

```

{
    "msg": "'NoneType' object has no attribute '__getitem__'",
    "data": "POST /yhs-robot/cmd/graph_check_point",
    "errCode": -1,
    "succeeded": false
}
// 地图名称错误

```

4.3.12 Verify If Multiple Line Segments Can Form a Path

Method: POST

URL: /cmd/graph_check_line

Body:

```

{
    "lines": [
        {
            "begin": {
                "x": 652,
                "y": 400
            }, // 开始点
            "end": {
                "x": 649,
                "y": 453
            }, // 结束点
            "radius": 0 // 半径
        },
        {
            "begin": {

```

```

        "x": 649,
        "y": 453
    }, // 开始点
    "end": {
        "x": 551,
        "y": 454
    }, // 结束点
    "radius": 0 // 半径
    }
],
"mapName": "0619-4" // 地图名称
}
// 需要将导航参数 is_check_obs_in_path 设置为 true

```

Response Example:

```

{
    "data": null,
    "errCode": 0,
    "msg": "路径可以通行",
    "succeeded": true
}

```

4.3.13 Start Recording Path

Method: GET

URL:

cmd/start_record_path?map_name={map_name}&path_name={path_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name
path_name	Yes	String	Recorded path name

Response Example:

```

{
    "data": null,
    "errCode": 0,
    "msg": "开始录制路径成功",
    "succeeded": true
}

```

Note:

- After the interface call is successful, you can use the remote control to move the car and record the trajectory. Note that recording a path only records the coordinates of the positions the car passes through, and does not record the car's movement mode (forward, backward, or lateral) or heading.
- When starting to record a path, try to keep the driving trajectory of the remote-controlled car smooth and with small curvature, otherwise the tracking linearity during navigation might be slightly poor.

4.3.14 Stop and Save Path

Method: GET

URL: /cmd/stop_record_path

Response Example:

```
{
  "data": {
    "coverageArea": null,
    "createAt": "2025-12-08 18:50:41",
    "data": [
      {
        "gridPosition": [],
        "length": 2.9473665990805413, // 路径长度
        "name": "__DEFAULT",
        "namedPoints": [
          {
            "actions": [],
            "angle": 2.5621489343398105, // 起点朝向角
            "gridPosition": {
              "x": 375,
              "y": 428
            },
            "index": 0, // 起点索引
            "moveType": null,
            "name": "__DEFAULT_0", // 起点名称
            "worldPose": {
              "orientation": {
                "w": 0.2856856877364595,
                "x": 0,
                "y": 0,
                "z": 0.9583233733049331
              },
              "position": {
                "x": 4.47639,
                "y": -0.97558599999999963,
                "z": 0
              }
            }
          }
        ]
      }
    ]
  }
}
```

```

    },
    {
      "actions": [],
      "angle": 3.0324743761406054, // 终点朝向角
      "gridPosition": {
        "x": 328,
        "y": 459
      },
      "index": 18, // 终点的索引
      "moveType": 1, // 导航模式 1 前进 2 后退 3 横移
      "name": "__DEFAULT_1", // 终点名称
      "worldPose": {
        "orientation": {
          "w": 0.05453207505835818,
          "x": 0,
          "y": 0,
          "z": 0.99851201935171
        },
        "position": {
          "x": 2.1263900000000024,
          "y": 0.5744140000000044,
          "z": 0
        }
      }
    }
  ],
  "pointCount": 19
},
{
  "fileName": "bd9dd698-d423-11f0-bd8a-c4837210cf9c.csv",
  "id": "bd9dd699-d423-11f0-bd8a-c4837210cf9c",
  "length": null,
  "mapId": "f225aa88-d3f8-11f0-bd8a-c4837210cf9c",
  "mapName": "map1",
  "name": "record", // 录制路径名称
  "pointCount": 19, // 录制路径点的数量
  "type": null
},
{
  "errCode": 0,
  "msg": "停止并保存路径成功",
  "successed": true
}
}

```

4.3.15 Cancel Path Recording

Method: GET

URL: /cmd/cancel_record_path

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "取消录制路径成功",
  "succeeded": true
}
```

4.3.16 Get Recorded Path

Method: GET

URL:
/data/record_path?map_name={map_name}&path_name={path_name}&contains_details=

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name
path_name	Yes	String	Recorded path name
contains_details	No	String	True or False (whether to include coordinates of all points in the path. Can be omitted if not needed, pass True if needed)

Response Example:

```
{
  "data": {
    "coverageArea": null,
    "createAt": "2024-10-31 17:55:00",
    "data": [
      {
        "gridPosition": [],
        "length": 6.457187620252355, // 录制路径长度
        "name": "__DEFAULT",
        "namedPoints": [
          {
            "actions": [], // 动作
            "angle": 1.567444871294638, // 角度
            "gridPosition": { // 录制路径起点栅格坐标
              "x": 142,
              "y": 342
            }
          }
        ]
      }
    ]
  }
}
```

```

        "index": 0, // 索引值
        "moveType": null, // 导航模式
        "name": "__DEFAULT_0", // 录制路径起点名称
        "worldPose": { // 世界坐标
            "orientation": {
                "w": 0.7082907062873957,
                "x": 0,
                "y": 0,
                "z": 0.7059208704854265
            },
            "position": {
                "x": 0.14688300000000076,
                "y": -0.45810699999999827,
                "z": 0
            }
        }
    },
    {
        "actions": [],
        "angle": 0.00825279705262194,
        "gridPosition": {
            "x": 256,
            "y": 358
        },
        "index": 30,
        "moveType": 1,
        "name": "__DEFAULT_1",
        "worldPose": {
            "orientation": {
                "w": 0.9999914864296812,
                "x": 0,
                "y": 0,
                "z": 0.0041263868161763185
            },
            "position": {
                "x": 5.8468830000000001,
                "y": 0.34189300000000244,
                "z": 0
            }
        }
    }
],
    "pointCount": 31 // 录制路径中点的个数
}

],
    "fileName": "31cfa224-976e-11ef-9db5-c4837210cf9c.csv", // 后端
中录制路径数据文件名称
    "id": "31cfa225-976e-11ef-9db5-c4837210cf9c",
    "length": null,

```

```

    "mapId": "8b0d382e-9755-11ef-9db5-c4837210cf9c", // 地图 ID
    "mapName": "mapa", // 地图名称
    "name": "a", // 录制路径名称
    "pointCount": 31, // 录制路径中路径点个数
    "type": null
  },
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}

```

4.3.17 Edit Action Point Data in Recorded Path

Method: PUT

URL: cmd/record_path_points

Body:

```

{
  "map_name": "map", // 地图名称
  "path_name": "a", // 录制路径名称
  "fileName": "31cfa224-976e-11ef-9db5-c4837210cf9c.csv", // 通过接口
“获取录制路径”获取
  "points": [
    {
      "actions": [],
      "angle": 0.0377617811628419,
      "gridPosition": {
        "x": 346,
        "y": 356
      },
      "index": 0, // 点的索引
      "name": "__DEFAULT_0",
      "worldPose": {
        "orientation": {
          "w": 0.999821761280518,
          "x": 0,
          "y": 0,
          "z": 0.018879768799504242
        },
        "position": {
          "x": 5.15,
          "y": -0.149636999999999847,
          "z": 0
        }
      }
    }
  ],
  {

```

```

    "actions": [
      {
        "fields": [
          {
            "fields": [],
            "name": "millisecond",
            "type": "3",
            "value": "3000"
          }
        ],
        "isChecked": false,
        "isDevice": false,
        "name": "Pause"
      },
      {
        "fields": [
          {
            "fields": [],
            "name": "rotation",
            "type": "double",
            "value": "90"
          }
        ],
        "isChecked": false,
        "isDevice": false,
        "name": "Rotate2d"
      }
    ],
    "angle": 0.22995333186748215,
    "gridPosition": {
      "x": 407,
      "y": 366
    },
    "index": 20,
    "name": "__DEFAULT_1",
    "worldPose": {
      "orientation": {
        "w": 0.9933974615506045,
        "x": 0,
        "y": 0,
        "z": 0.1147235084227089
      },
      "position": {
        "x": 8.200000000000001,
        "y": 0.35036300000000153,
        "z": 0
      }
    }
  }
}

```



```
]
}
```

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "编辑录制路径动作点成功",
  "successed": true
}
```

Note:

- You can obtain the grid coordinates of each point in the recorded path using the “Get Recorded Path” interface, and then use this interface to set actions for a specific point based on its index.

4.3.18 Delete Recorded Path

Method: DELETE

URL: /cmd/record_path?map_name={map_name}&path_name={path_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name
path_name	Yes	String	Recorded path name

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "删除录制路径成功",
  "successed": true
}
```

4.3.19 Get Recorded Path List

Method: GET

URL: /data/record_list?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name

4.3.20 Rename Recorded Path

Method: PUT

URL:

/cmd/update_rec_path_name?map_name={map_name}&origin_path_name={origin_path_name}&new_path_name={new_path_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name
origin_path_name	Yes	String	Old recorded path name
new_path_name	Yes	String	New recorded path name

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "重命名路径成功",
  "succeeded": true
}
```

4.3.21 Validate Hand-Drawn Path Name

Method: GET

URL: /cmd/graph_check_path_name?map_id={map_id}&name={path_name}

Query:

Parameter	Required	Type	Description
map_id	Yes	String	Map ID
path_name	Yes	String	Hand-drawn path name

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "验证手绘路径名称成功",
  "succeeded": true
}
```

4.4 Task Module

4.4.1 Start Executing Task

Method: POST

URL: /task_queue/start

Body:

```
{
  "loop": false, // 是否循环执行
  "loop_count": 0, // 循环次数, 当 loop 为 false 时, loop_count 默认为 1
  "map_name": "map1", // 地图名称
  "name": "", // 路径组合名称, 可选, 需要执行路径组合任务时, 传入该值。当
  对应的队列数据不存在时, 则执行 tasks 中的数据。
  "tasks": [
    {
      "name": "NavigationPointTask", // 导航点任务
      "start_param": {
        "map_name": "map1", // 地图名称
        "position_name": "goal" // 导航点名称
      }
    },
    {
      "name": "NavigationPositionTask", // 根据位姿下发导航点
      "start_param": {
        "destination": {
          "angle": 0, // 导航的角度 (范围: -180°~180°)
          "gridPosition": { // 导航点栅格坐标
            "x": 170,
            "y": 396
          }
        }
      }
    },
    {
      "name": "PlayGraphPathTask", // 手绘路径任务
      "start_param": {
        "graph_name": "path1", // 手绘路径名称
        "map_name": "" // 地图名称
      }
    },
    {
      "name": "PlayPathTask", // 录制路径任务
      "start_param": {
        "map_name": "map1", // 地图名称
        "path_name": "record" // 录制路径名称
      }
    }
  ]
}
```

```
}  
}  
]  
}
```

Response Example:

```
{  
  "data": "97a9dfbe-d427-11f0-bd8a-c4837210cf9c",  
  "errCode": 0,  
  "msg": "成功",  
  "succeeded": true  
}
```

Failure Example:

```
{ // Body 内容错误  
  "msg": "too many values to unpack",  
  "data": "POST /yhs-robot/task_queue/start",  
  "errCode": -1,  
  "succeeded": false  
}
```

Note:

- When there are multiple tasks in tasks, they are executed in order.
- Currently, tasks cannot be preempted after execution; the currently running task must be stopped first.

4.4.2 Pause Task

Method: POST

URL: /task_queue/pause

Response Example:

```
{  
  "data": null,  
  "errCode": 0,  
  "msg": "成功",  
  "succeeded": true  
}
```

Failure Example:

```
{  
  "data": null,  
  "errCode": -1,  
  "msg": "没有运行中的任务队列",  
}
```

```
    "succeeded": false
}
```

4.4.3 Resume Task

Method: POST

URL: /task_queue/resume

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "成功",
  "succeeded": true
}
```

Failure Example:

```
{
  "data": null,
  "errCode": -1,
  "msg": "没有已暂停的任务队列",
  "succeeded": false
}
```

4.4.4 Stop All Tasks

Method: POST

URL: /task_queue/stop

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "成功",
  "succeeded": true
}
```

Failure Example:

```
{
  "data": null,
  "errCode": -1,
  "msg": "没有运行中的任务队列",
  "succeeded": false
}
```

4.4.5 Stop Current Task / Skip Current Task

Method: POST

URL: /task_queue/stop/task

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "成功",
  "succeeded": true
}
```

Failure Example:

```
{
  "data": null,
  "errCode": -1,
  "msg": "没有运行中的任务队列",
  "succeeded": false
}
```

Note:

- If there is only one task, it will stop the task. If there are multiple tasks, it will skip the current task.
- Task combinations of path points can be manually skipped via the interface, but if there is an obstacle at a target point, navigation will end; for hand-drawn paths, if there is an obstacle at a point, the current point is automatically skipped.

4.4.6 Query If Path Combination Exists

Method: GET

URL:

/task_queue/exist?map_name={map_name}&queue_name={queue_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name
queue_name	Yes	String	Path combination name

Response Example:

```
{
  "data": false, // 路径组合不存在
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Response Example 2:

```
{
  "data": true, // 路径组合存在
  "errCode": -1,
  "msg": "error",
  "succeeded": false
}
```

4.4.7 Add Save Path Combination

Method: POST

URL: /task_queue

Body:

```
{
  "name": "combined", // 路径组合名称
  "map_name": "map", // 地图名称
  "loop": false,
  "tasks": [
    {
      "name": "PlayPathTask", // 录制路径
      "start_param": {
        "path_name": "record" // 录制路径名称
      }
    },
    {
      "name": "PlayGraphPathTask", // 手绘路径
      "start_param": {
        "graph_name": "graph" // 手绘路径名
      }
    },
    {
      "name": "NavigationPointTask", // 导航到点
      "start_param": {
        "position_name": "goal" // 点名称
      }
    },
    {
      "name": "NavigationPositionTask", // 导航到具体位置
    }
  ]
}
```

```

        "start_param": {
            "destination": {
                "angle": 0, // 点的角度
                "gridPosition": { // 点的栅格坐标
                    "x": 265,
                    "y": 371
                }
            }
        }
    ]
}

```

Response Example:

```

{
    "data": null,
    "errCode": 0,
    "msg": "成功",
    "succeeded": true
}

```

4.4.8 Update Path Combination

Method: PUT

URL: /task_queue

Body:

```

{
    "name": "combined",
    "map_name": "map",
    "loop": true,
    "tasks": [
        {
            "name": "PlayPathTask", // 任务类型
            "start_param": {
                "path_name": "record" // 录制路径名称
            }
        },
        {
            "name": "PlayGraphPathTask",
            "start_param": {
                "graph_name": "graph" // 手绘路径名称
            }
        }
    ]
}

```


Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "成功",
  "succeeded": true
}
```

4.4.9 Get Path Combination List

Method: GET

URL: /task_queue/list?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name

Response Example:

```
{
  "data": [
    {
      "createAt": "2024-11-01 17:03:29", // 创建时间
      "loop": true,
      "map_id": "8b0d382e-9755-11ef-9db5-c4837210cf9c", // 地图 ID
      "map_name": "map", // 地图名称
      "name": "combined", // 路径组合名称
      "tasks": [
        {
          "name": "PlayPathTask", // 录制路径
          "start_param": {
            "path_name": "record" // 录制路径名称
          }
        },
        {
          "name": "PlayGraphPathTask", // 手绘路径
          "start_param": {
            "graph_name": "graph" // 手绘路径名称
          }
        },
        {
          "name": "NavigationPointTask", // 导航点
          "start_param": {
            "position_name": "goal" // 导航点名称
          }
        }
      ]
    }
  ]
}
```

```

    },
    {
        "name": "NavigationPositionTask",
        "start_param": {
            "destination": {
                "angle": 0,
                "gridPosition": {
                    "x": 265,
                    "y": 371
                }
            }
        }
    }
]
},
{
    "createAt": "2024-11-01 17:17:40",
    "loop": false,
    "map_id": "8b0d382e-9755-11ef-9db5-c4837210cf9c",
    "map_name": "maph",
    "name": "c2",
    "tasks": [
        {
            "key": "1", // 执行顺序
            "name": "NavigationPointTask",
            "start_param": {
                "map_name": "maph",
                "position_name": "goal"
            }
        }
    ]
}
],
"errCode": 0,
"msg": "success",
"successed": true
}

```

4.4.10 Delete Task Queue

Method: DELETE

URL:

/task_queue?map_name={map_name}&task_queue_name={task_queue_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name

Parameter	Required	Type	Description
task_queue_name	Yes	String	Path combination name

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "成功",
  "succeeded": true
}
```

4.4.11 Whether the Current Task Is Complete

Method: GET

URL: /task_queue/task/is_finished

Response Example:

```
// 完成/无法到达/取消
{
  "data": true, // 当前没有任务在运行中
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Response Example 2:

```
{
  "data": false, // 当前有任务正在运行中
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Note:

- When the task is completed, unreachable, or cancelled, this interface returns true, indicating that no task is currently running. It is mainly used to determine if a task is currently in progress.

4.4.12 Whether the Current Queue Execution Is Complete

Method: GET

URL: /task_queue/is_finished

Response Example:

```
{
  "data": 1, // 任务正在执行中
  "errCode": 0,
  "msg": "",
  "succeeded": true
}
```

Response Example 2:

```
{
  "data": 0, // 没有任务正在运行中
  "errCode": 0,
  "msg": "",
  "succeeded": true
}
```

Failure Example:

```
{
  "data": 0,
  "errCode": 0,
  "msg": "不能到达", // 导航无法到达目标点
  "succeeded": true
}
```

Note:

- If data returns 0 and msg outputs “不能到达”, the car or target point might be inside an obstacle.

4.4.13 Query Task Queue Details

Method: GET

URL:

/task_queue/detail?map_name={map_name}&queue_name={queue_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name
queue_name	Yes	String	Path combination name

Response Example:

```
{
  "data": {
```

```

    "loop": false,
    "map_name": "map4",
    "name": "group2",
    "tasks": [
      {
        "name": "NavigationPointTask",
        "start_param": {
          "map_name": "map4",
          "position_name": "init1"
        }
      },
      {
        "name": "PlayGraphPathTask",
        "start_param": {
          "graph_name": "path4",
          "map_name": "map4"
        }
      }
    ]
  },
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}

```

4.4.14 Get Task Log/Report

Method: GET

URL:

/task_queue/log?start_time={start_time}&end_time={end_time}&map_name={map_name}

Query:

Parameter	Required	Type	Description
start_time	Yes	String	Start time
end_time	Yes	String	End time
map_name	Yes	String	Map name

Response Example:

```

{
  "data": [
    {
      "_sa_instance_state": {
        "class_": {
          "py/type": "database.models.tb_task_log.TaskLog"
        }
      }
    }
  ]
}

```

```

        "committed_state": {},
        "expired_attributes": [],
        "instance": null,
        "key": [
            {
                "py/type": "database.models.tb_task_log.TaskLog
"
            },
            [
                13478
            ],
            null
        ],
        "load_options": [],
        "load_path": [
            {
                "py/type": "database.models.tb_task_log.Tas
kLog"
            },
            null
        ],
        "manager": {
            "class_": {
                "py/type": "database.models.tb_task_log.TaskLog
"
            }
        },
        "endTime": "2025-12-08T19:18:51.297655",
        "executor": "admin",
        "executorCode": "admin",
        "id": 13478,
        "is_success": true,
        "map_name": "map1",
        "name": null,
        "remark_2": "97a9dfbe-d427-11f0-bd8a-c4837210cf9c",
        "remark_3": null,
        "startTime": "2025-12-08T19:18:16.178683",
        "taskLogType": "NavigationPositionTask",
        "task_data": "{\"map_name\\\": \\\"map1\\\", \\\"type\\\": null, \\\"ta
sks\\\": [{\\\"start_param\\\": {\\\"map_name\\\": \\\"map1\\\", \\\"position_name\\\": \\
\"init\\\", \\\"name\\\": \\\"NavigationPointTask\\\"}, {\\\"start_param\\\": {\\\"dest
ination\\\": {\\\"angle\\\": -120, \\\"gridPosition\\\": {\\\"y\\\": 447, \\\"x\\\": 28
5}}}, \\\"name\\\": \\\"NavigationPositionTask\\\"}], \\\"loop_count\\\": 1, \\\"loop
\\\": false}\"
    },
    {
        "_sa_instance_state": {

```

```

        "class_": {
            "py/type": "database.models.tb_task_log.TaskLog"
        },
        "committed_state": {},
        "expired_attributes": [],
        "instance": null,
        "key": [
            {
                "py/type": "database.models.tb_task_log.TaskLog"
            },
            [
                13477
            ],
            null
        ],
        "load_options": [],
        "load_path": [
            {
                "py/type": "database.models.tb_task_log.Tas
kLog"
            },
            null
        ],
        "manager": {
            "class_": {
                "py/type": "database.models.tb_task_log.TaskLog"
            }
        },
        "endTime": "2025-12-08T19:17:19.718191",
        "executor": "admin",
        "executorCode": "admin",
        "id": 13477,
        "is_success": true,
        "map_name": "map1",
        "name": null,
        "remark_2": "6e8051fe-d427-11f0-bd8a-c4837210cf9c",
        "remark_3": null,
        "startTime": "2025-12-08T19:17:07.119445",
        "taskLogType": "NavigationPositionTask",
        "task_data": "{\\"map_name\\": \\"map1\\", \\"type\\": null, \\"ta
sks\\": [{\\"start_param\\": {\\"destination\\": {\\"angle\\": -120, \\"gridPos
ition\\": {\\"y\\": 447, \\"x\\": 285}}}, \\"name\\": \\"NavigationPositionTask
\\"}], \\"loop_count\\": 1, \\"loop\\": false}"
    },
],

```

```
"errCode": 0,
"msg": "success",
"successed": true
}
```

4.4.15 Get Currently Executing Path Combination + Current Status

Method: GET

URL: /task_queue/current

Response Example:

```
{ // 执行任务中:
  "data": {
    "data": {
      "group_name": null, // 路径组合名称
      "loop": false,
      "loop_count": 1,
      "map_name": "in",
      "tasks": [
        {
          "name": "PlayGraphPathTask",
          "start_param": {
            "graph_name": "graph"
          }
        }
      ],
      "type": "show_draw" // show_draw 手绘路径, show_group 路径组合,
show_record 录制路径
    },
    "status": 1 // 1 运行中 2 暂停中
  },
  "errCode": 0,
  "msg": "success",
  "successed": true
}
```

Response Example 2:

```
{ // 未执行任务:
  "data": {
    "data": null,
    "status": 0
  },
  "errCode": 0,
  "msg": "success",
  "successed": true
}
```


4.4.16 Edit Task Queue

Method: PUT

URL: /task_queue

Body :

```
{
  "name": "combined",
  "map_name": "in",
  "loop": true,
  "tasks": [
    {
      "name": "NavigationPositionTask",
      "start_param": {
        "destination": {
          "angle": 0,
          "gridPosition": {
            "x": 265,
            "y": 371
          }
        }
      }
    }
  ]
}
```

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "成功",
  "succeeded": true
}
```

4.4.17 Upload Task Image

Method: POST

URL: /task_queue/upload_task_image

Body :

[illegible]

```
AQEBAQEAAAAAAAAAAECAwQFBgcICQoL/8QAtRAAAgEDAwIEAwUFBAQAAAF9AQIDAAQRBRI
hMUEGE1FhByJxFDKBkaEII0KxwRVS0fAkM2JyggkKFhcYGRolJicoKSo0NTY....."
}
```

Response Example:

```
{
  "data": "上传成功",
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Failure Example:

```
{
  "msg": "list index out of range",
  "data": "POST /yhs-robot/task_queue/upload_task_image",
  "errCode": -1,
  "succeeded": false
}
```

Note:

- Upload a base64 format image to the backend. The base64 format input needs to be in the standard format, with the data header::data:image/jpeg;base64,

4.4.18 Get Task Report Image

Method: GET

URL: /task_queue/down_task_image?task_id={task_id}

Query:

Parameter	Required	Type	Description
task_id	Yes	String	Task ID

Response Example:

```
{
  "data": "/9j/4AAQSkZJRgABAQAAQABAAQ/4gIoSUNDX1BST0ZJTEUAAQEAABIAAA
AAAAIQAABtbnRyUkdCIFhZWiAAAAAAAAAAAAAAAAABhY3NwAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAQAA9tYAAQAAAADTLQAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAkZXNjAAAA8AAAAHRYWFlaAAABZAAAABRnWFlaAAABeAA
AABRiWFlaAAABjAAAABRyVFJDAAABoA.....",
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Note:

- How to get the task_id in the URL: The remark_2 field in the data returned by the interface (Get Task Log/Report – /task_queue/log) is the task_id.

4.4.19 Rename Path Combination

Method: PUT**URL:** /task_queue/rename**Body:**

```
{
  "map_name": "in",
  "old_name": "combined",
  "new_name": "combine"
}
```

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "成功",
  "succeeded": true
}
```

Failure Example:

```
{
  "data": null,
  "errCode": -1,
  "msg": "任务数据不存在",
  "succeeded": false
}
```

4.5 Autonomous Charging Module

4.5.1 Autonomous Recharging

Method: POST**URL:** /cmd/recharge**Body:**

```
{
  "map_name": "map", //地图名称
  "point_name": "recharge" //充电点名称
}
```

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "开始充电...",
  "succeeded": true
}
```

Response Example:

```
{
  "data": null,
  "errCode": -1,
  "msg": "已经在去往充电点",
  "succeeded": false
}
```

Response Example 2:

```
{
  "data": null,
  "errCode": -1,
  "msg": "正在充电中...",
  "succeeded": false
}
```

Note:

- When executing recharging, the car first goes to a temporary point in front of the charging pile, then docks backwards. Ensure the area in front of the charging pile is clear and free of obstacles.
- Due to its unique motion model, an Ackermann car has a larger movement range when adjusting posture, so ensure the area around the charging pile is sufficiently open.

4.5.2 Charging Status Feedback

Method: GET

URL: /data/recharge/status

Response Example:

```
{
  "data": 1, // 请先下发充电指令
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Response Example 2:

```
{
  "data": 2, // 正在去往充电桩
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Response Example 3:

```
{
  "data": 3, // 充电失败
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Response Example 4:

```
{
  "data": 4, // 充电成功
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

4.5.3 Disconnect from Charging Pile

Method: POST

URL: /cmd/dis_recharge

Response Example:

```
{
  "data": 2,
  "errCode": 0,
  "msg": "未充电，不需要脱离",
  "succeeded": true
}
```

Response Example 2:

```
{
  "data": 1,
  "errCode": 0,
  "msg": "开始脱离",
  "succeeded": true
}
```

4.5.4 Disconnect from Charging Pile Result Feedback

Method: GET

URL: /cmd/dis_recharge/status

Response Example:

```
{
  "data": 1, // 请先下发脱离指令
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Response Example 2:

```
{
  "data": 2, // 正在脱离
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Response Example 3:

```
{
  "data": 4, // 脱离成功
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Response Example 4:

```
{
  "data": 3, // 脱离失败
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Note:

- When disconnecting from the pile, ensure there are no obstacles in front of the car.

4.5.5 Pause Autonomous Recharging

Method: POST

URL: /cmd/recharge/pause

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "成功",
  "succeeded": true
}
```

Note:

- This interface only pauses the task of the car moving to the temporary point of the charging pile. It will not pause when the car has already started backing up from the temporary point for recharging.

4.5.6 Resume Autonomous Recharging

Method: POST

URL: /cmd/recharge/resume

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "成功",
  "succeeded": true
}
```

4.5.7 Cancel Autonomous Recharging

Method: POST

URL: /cmd/recharge/cancel

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "成功",
  "succeeded": true
}
```

4.6 Real-time Data

4.6.1 Robot's Real-time Position on Map

Method: GET

URL: /real_time_data/robot_pos

Response Example:

```
{
  "data": {
    "angle": 264.41268743865476, // 机器人朝向角度 (°)
    "gridPosition": { // 栅格坐标
      "x": 225,
      "y": 563
    },
    "mapInfo": null,
    "worldPosition": { // 世界坐标
      "position": {
        "x": -0.3090598498382481,
        "y": 1.2287962291971526
      }
    }
  },
  "errCode": 0,
  "msg": "数据获取成功",
  "succeeded": true
}
```

Note:

- When getting the robot's position, ensure the car's localization is accurate.

4.6.2 Get Robot Status Data

Method: GET

URL: /real_time_data/robot_hardware_status

Response Example:

```
{
  "data": {
    "anticollision": 0, // 防撞条状态 0: 没触发 1: 前防撞条触发 2: 后防撞条触发 3: 前后防撞条触发
    "battery_percentage": 66, // 电量百分比(bms_flag_fb_soc)
    "camera_num": 1, // 相机数量
    "cameras": 0, // 相机状态 cameras_status (未使用)
    "cameras_obs": 0, // ros_backend_interface (未使用)
    "charge": 0, // 充电状态 (充电时为 1, 未充电时为 0) charge_status
    "current_map_name": "", // 建图时, 会显示当前正在建图的地图名称
    "current_working": 3, // 1 未运行, 2 扫地图, 3 导航启动, 4 正在执行导航 (刚启动导航时是 3, 执行导航中是 4, 执行完导航任务后为 1)
    "emergency_stop": false, // 急停按钮状态 0: 没触发 1: 触发
  }
}
```



```

"error": 0, // 故障代码, APP 上会出现警告标志
"gps": false, // gps 状态 gps_status (未使用)
"gps_num": 0, // gps 数量
"header": {
    "frame_id": "",
    "seq": 7194,
    "stamp": {
        "nsecs": 0,
        "secs": 0
    }
},
"imu": true, // imu 状态
"io": 0, // io 状态
"lidar": true, // 雷达状态, 订阅的话题是/rslidar_points
"lidar_num": 1,
"map_name": "map", // 当前正在使用的默认地图
"odom": true, // odom 状态
"recharge_point": { // 执行充电任务时显示, 其他情况为 null
    "angle": 0,
    "createdAt": "2024-08-06 10:07:27",
    "gridX": 280,
    "gridY": 414,
    "id": "c0784266-5398-11ef-84fa-c4837210cf83", // 点 ID
    "mapId": "13578218-5302-11ef-8e00-c4837210cf83",
    "mapName": "map",
    "markPoint": false, // 标记当前点
    "name": "a", // 充电点名称
    "type": 1, // 初始化点:0, 充电点:1, 导航点:2, RFID 点:3, 注水
点:4, 排水点:5, 二维码点:11, 导航结束点:12
    "worldPose": { // 充电点世界坐标
        "orientation": {
            "w": 1,
            "x": 0,
            "y": 0,
            "z": 0
        },
        "position": {
            "x": -0.57218300000000008,
            "y": 2.1000000000000014,
            "z": 0
        }
    }
},
"recharge_res": 1, // 和接口“充电状态反馈”一致
"theta": 0.041458728252612036, // 以 map 的 x 坐标为准, 左边 0°~18
0°, 右边 0°~-180°, 再转换成弧度
"type": 1, // 底盘类型: 1 差速底盘, 2 四转四驱底盘, 3 阿克曼底盘

```

```

    "ultrasonic": 1, // 超声波状态, ultrasonic_status 变量
    "ultrasonic_get_obs": 0, // ros_backend_interface (未使用)
    "ultrasonic_num": 0, // 超声波数量
    "world_x": -1.729352705370178, // 机器人当前位置世界坐标
    "world_y": 0.00015473257010497265
  },
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}

```

4.6.3 Get Real-time Linear Velocity/Angular Velocity

Method: GET

URL: /real_time_data/angular_linear

Response Example:

```

{
  "data": {
    "angle": 1.6472212278609724, // 机器人朝向角度 0°~360°
    "angular": { // 角速度 rad/s
      "x": 0,
      "y": 0,
      "z": 0.0031400001607835293
    },
    "linear": { // 线速度 m/s
      "x": 0,
      "y": 0,
      "z": 0
    }
  },
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}

```

4.6.4 Get Odometry Data

Method: GET

URL: /real_time_data/odom

Response Example:

```

{
  "data": {
    "child_frame_id": "base_link",
    "header": {

```

```
    "frame_id": "odom",
    "seq": 18249,
    "stamp": {
        "nsecs": 610916215,
        "secs": 1684893272
    }
},
"pose": {
    "covariance": [
        0.1,
        0,
        0,
        .
        .
        .
        0,
        0,
        0.2
    ],
    "pose": { // 机器人位姿
        "orientation": {
            "w": 1,
            "x": 0,
            "y": 0,
            "z": 0
        },
        "position": {
            "x": 0,
            "y": 0,
            "z": 0
        }
    }
},
"twist": {
    "covariance": [
        0,
        0,
        .
        .
        .
        0,
        0
    ],
    "twist": { // 速度
        "angular": {
            "x": 0,
            "y": 0,
            "z": 0
        },
    },
}
```

```

        "linear": {
            "x": 0,
            "y": 0,
            "z": 0
        }
    },
    "errCode": 0,
    "msg": "success",
    "succeeded": true
}

```

4.6.5 Get IMU Data

Method: GET

URL: /real_time_data/imu

Response Example:

```

{
    "data": {
        "angular_velocity": {
            "x": 0,
            "y": 0,
            "z": 0.0017453290260074287 // Z 轴角速度
        },
        "header": {
            "frame_id": "imu_link",
            "seq": 501486,
            "stamp": {
                "nsecs": 857216262,
                "secs": 1730444438
            }
        },
        "linear_acceleration": { // 线加速度
            "x": 0.08819999620318414,
            "y": 0.31360001489520073,
            "z": 9.809800457954408
        },
        "orientation": {
            "w": 0.9998740553855896,
            "x": 0.015088269487023354,
            "y": -0.00487076910212636,
            "z": 0.0008079399704001844
        }
    },
    "errCode": 0,
    "msg": "success",
}

```

```
    "succeeded": true
}
```

4.6.6 Get Robot Localization Status

Method: GET

URL: /real_time_data/robot_local_status

Response Example:

```
{
  "data": true, // 定位成功
  "errCode": 0,
  "msg": "数据获取成功",
  "succeeded": true
}
```

Response Example 2:

```
{
  "data": false, // 定位失败
  "errCode": 0,
  "msg": "数据获取成功",
  "succeeded": true
}
```

4.6.7 Get 2D Laser Scan Data

Method: GET

URL: /real_time_data/laser2d/scan

Response Example:

```
{
  "data": {
    "angle_increment": 0.008700000122189522,
    "angle_max": 3.140000104904175,
    "angle_min": -3.140000104904175,
    "header": {
      "frame_id": "laser_link",
      "seq": 16391,
      "stamp": {
        "nsecs": 703792572,
        "secs": 1765260283
      }
    },
    "intensities": [],
    "range_max": 50,
    "range_min": 0.20000000298023224,
  }
}
```

```

    "ranges": [
      null,
      null,
      2.1740684509277344,
      1.9068375825881958,
      .
      .
      .
      2.174074172973633,
      1.852486491203308,
      2.1640944480895996,
      null,
      null
    ],
    "scan_time": 0.33329999446868896,
    "time_increment": 0
  },
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}

```

Note:

- The scan data obtained by this interface is in the world coordinate system.

4.6.8 Get 2D Laser Grid Data

Method: GET

URL: /real_time_data/las_gril

Response Example:

```

{
  "data": {
    "gridPhits": [
      {
        "x": 255,
        "y": 479
      },
      {
        "x": 254,
        "y": 479
      },
      {
        "x": 257,
        "y": 474
      },
      {
        "x": 254,

```

```

        "y": 478
      },
      .
      .
      {
        "x": 428,
        "y": 438
      }
    ],
    "header": null,
    "mapInfo": null,
    "worldPhits": null
  },
  "errCode": 0,
  "msg": "数据获取成功",
  "succeeded": true
}

```

Note:

- The laser scan data obtained by this interface is in the grid coordinate system. The red laser data displayed on the APP is based on this interface.

4.6.9 Get 3D Laser Data

Method: GET

URL: /real_time_data/laser3d/lidar_cloud

Response Example:

```

{
  "data": {
    "header": {
      "frame_id": "laser_link",
      "seq": 906,
      "stamp": {
        "nsecs": 104513645,
        "secs": 1765260863
      }
    },
    "poses": [
      {
        "position": {
          "x": 14.354354858398438,
          "y": -0.8281758427619934,
          "z": 3.3087761402130127
        }
      }
    ]
  }
}

```

```

    {
      "position": {
        "x": 14.336942672729492,
        "y": -0.8748798370361328,
        "z": 3.3054044246673584
      }
    },
    .
    .
    .
    {
      "position": {
        "x": 2.4174299240112305,
        "y": 0.3238258957862854,
        "z": -0.5546470880508423
      }
    }
  ]
},
"errCode": 0,
"msg": "success",
"successed": true
}

```

Note:

- The amount of lidar point cloud data is very large; it is not recommended to obtain data from the interface for use.

4.6.10 Get GPS Data

Method: GET

URL: /real_time_data/gps

Response Example:

```

{
  "data": {
    "altitude": 77,
    "header": {
      "frame_id": "",
      "seq": 33,
      "stamp": {
        "nsecs": 34,
        "secs": 40
      }
    }
  },
  "latitude": 30,
  "longitude": 88,
  "service": 59,

```



```

        "status": 16,
        "yaw": 0
    },
    "errCode": 0,
    "msg": "success",
    "succeeded": true
}

```

Note:

- When obtaining data, the first call to the interface may return empty data. Because after the topic data is monitored, for performance, the topic data is cached and refreshed in real-time, so the latest data is obtained on the next request. Normally, the topic should always exist, or it will refresh the previous data when the topic monitoring succeeds next time. Therefore, currently, the cache is not cleared by monitoring whether the topic has disappeared.

4.6.11 Get Raw Ultrasonic Data

Method: GET

URL: /real_time_data/ultrasonic

Response Example:

```

{
  "data": {
    "front_left": 1390, // 前左超声波 mm
    "front_right": 1040, // 前右
    "header": {
      "frame_id": "",
      "seq": 2174,
      "stamp": {
        "nsecs": 0,
        "secs": 0
      }
    },
    "left_front": 1390, // 左前
    "left_rear": 1470, // 左后
    "rear_left": 1580, // 后左
    "rear_right": 1680, // 后右
    "right_front": 520, // 右前
    "right_rear": 440 // 右后
  },
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}

```

4.6.12 Get Ultrasonic Encapsulated Data/scan3

Method: GET

URL: /real_time_data/laser2d/scan3

Response Example:

```
{
  "data": {
    "angle_increment": 0.17499999701976776,
    "angle_max": 3.140000104904175,
    "angle_min": -3.140000104904175,
    "header": {
      "frame_id": "ul_link",
      "seq": 2028,
      "stamp": {
        "nsecs": 360195903,
        "secs": 1734750787
      }
    },
    "intensities": [
      0,
      0,
      0,
      .
      .
      .
      0,
      0,
      0
    ],
    "range_max": 5,
    "range_min": 0,
    "ranges": [
      null,
      null,
      null,
      .
      .
      .
      1.1341135501861572,
      null,
      null
    ],
    "scan_time": 0.10000000149011612,
    "time_increment": 0
  },
  "errCode": 0,
  "msg": "success",
}
```

```
    "succeeded": true
}
```

4.6.13 Get Chassis and Navigation Version Information

Method: GET

URL: /real_time_data/chassis_navigation

Response Example:

```
{
  "data": {
    "chassis_version": "1.0.0", // 底盘版本
    "header": {
      "frame_id": "",
      "seq": 736,
      "stamp": {
        "nsecs": 0,
        "secs": 0
      }
    },
    "navigation_version": "1.0.1", // 导航版本
    "type": 3
  },
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

4.6.14 Get Estimated Task Execution Time

Method: GET

URL: /real_time_data/task_process_status

Response Example:

```
{
  "data": {
    "errCode": 0,
    "statusCode": 1, // 1 导航中
                    // 2 导航取消
                    // 3 到达目标点（未运行）
                    // 4 不能到达
    "statusData": {
      "less_second": 1, // 剩余时间(s)
      "loop_count": 1, // 总循环次数
      "loop_times": 1, // 当前在第几次循环
    }
  }
}
```

```

        "process_rate": 0.5584237670971816 // 当前进度 百分比
    },
    "statusMsg": "following path"
},
"errCode": 0,
"msg": "数据获取成功",
"successed": true
}

```

4.6.15 Get Real-time Navigation Local Path Route Data

Method: GET

URL: /real_time_data/local_plan_path

Response Example:

```

{
  "data": {
    "gridPhits": [ // 局部路径上每个点的世界坐标
      {
        "x": 0.02376289203797821,
        "y": -0.298854370275338
      },
      {
        "x": 0.09612715273307905,
        "y": -0.2983048704951583
      },
      {
        "x": 0.16058310332728803,
        "y": -0.2927794298508405
      },
      {
        "x": 0.229661403397927,
        "y": -0.28276703581023355
      }
    ]
  },
  "errCode": 0,
  "msg": "数据获取成功",
  "successed": true
}

```

4.6.16 Get Real-time Navigation Global Path Route Data Grid Coordinates

Method: GET

URL: /real_time_data/local_grid_plan_path

Response Example:

```

{
  "data": {
    "gridPhits": [ // 全局路径上每个点的栅格坐标
      {
        "x": 141,
        "y": 345
      },
      {
        "x": 157,
        "y": 345
      },
      {
        "x": 157,
        "y": 345
      }
    ]
  },
  "errCode": 0,
  "msg": "数据获取成功",
  "succeeded": true
}

```

4.6.17 Get Depth Camera Point Cloud Converted Data scan1

Method: GET

URL: /real_time_data/laser2d/scan1

Response Example:

```

{
  "data": {
    "angle_increment": 0.008700000122189522,
    "angle_max": 3.140000104904175,
    "angle_min": -3.140000104904175,
    "header": {
      "frame_id": "ascamera_hp60c_ln_1", // 前相机
      "seq": 12921,
      "stamp": {
        "nsecs": 229472162,
        "secs": 1765261815
      }
    },
    "intensities": [],
    "range_max": 2,
    "range_min": 0.10000000149011612,
    "ranges": [
      null,
      null,
      null,

```

```

        .
        .
        .
        null,
        null,
        null
    ],
    "scan_time": 0.33329999446868896,
    "time_increment": 0
},
"errCode": 0,
"msg": "success",
"successed": true
}

```

4.6.18 Get Depth Camera Point Cloud Converted Data scan2

Method: GET

URL: /real_time_data/laser2d/scan2

Response Example:

```

{
  "data": {
    "angle_increment": 0.008700000122189522,
    "angle_max": 3.140000104904175,
    "angle_min": -3.140000104904175,
    "header": {
      "frame_id": "ascamera_hp60c_ln_2", // 后相机
      "seq": 232215,
      "stamp": {
        "nsecs": 522402921,
        "secs": 1731392298
      }
    },
    "intensities": [],
    "range_max": 2,
    "range_min": 0.10000000149011612,
    "ranges": [
      null,
      null
    ],
    "scan_time": 0.33329999446868896,
    "time_increment": 0
  },
  "errCode": 0,
  "msg": "success",
  "successed": true
}

```

4.6.19 Front Camera Point Cloud Data

Method: GET

URL: /real_time_data/front_camera

Response Example:

```
{
  "data": {
    "header": {
      "frame_id": "ascamera_hp60c_ln_1",
      "seq": 301,
      "stamp": {
        "nsecs": 455535706,
        "secs": 1731287435
      }
    },
    "poses": [
      {
        "position": {
          "x": 1.246752381324768,
          "y": 0.21409708261489868,
          "z": -0.9743396639823914
        }
      },
      {
        "position": {
          "x": 1.24318265914917,
          "y": -0.31266897916793823,
          "z": -0.9715498685836792
        }
      }
    ]
  },
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Note:

- It is not recommended to obtain point cloud data from the interface.

4.6.20 Rear Camera Point Cloud Data

Method: GET

URL: /real_time_data/rear_camera

Response Example:

```
{
  "data": {
    "header": {
      "frame_id": "ascamera_hp60c_1n_2",
      "seq": 301,
      "stamp": {
        "nsecs": 455535706,
        "secs": 1731287435
      }
    },
    "poses": [
      {
        "position": {
          "x": 1.246752381324768,
          "y": 0.21409708261489868,
          "z": -0.9743396639823914
        }
      },
      {
        "position": {
          "x": 1.24318265914917,
          "y": -0.31266897916793823,
          "z": -0.9715498685836792
        }
      }
    ]
  },
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

Note:

- It is not recommended to obtain point cloud data from the interface.

4.6.21 Front Camera Captured Color Image in Base64 Format

Method: GET

URL: /real_time_data/ahl1_rgb_img

Response Example:

```
{
  "data": "/9j/4AAQSkZJRgABAQAAQABAAD/2wBDAAIBAQEBAQIBAQECAGICAgQDAG  
ICAgUEBAMEBgUGBgYFBgYGBWkIBGcJBWYGCAsICQoKCgoKBggLDAsKDAKKGgr/2wBDAQICA  
gICAgUDAwUKBwYHCgoKcGoKcGoKcGoKcGoKcGoKcGoKcGoKcGoKcGoKcGoKcGoKcGoKcGo  
K.....Urz7P/z71zeu/Zv0/cV0F5XN6xCf061EJczAiS+v4Vt2c/7nNYFnBwjDBxXS3ymZ  
uef7VRvJ6IO34VFL0/Co5Be93P/Z",
  "errCode": 0,
```



```
"msg": "success",
"successed": true
}
```

4.6.22 Rear Camera Captured Color Image in Base64 Format

Method: GET

URL: /real_time_data/ahl2_rgb_img

Response Example:

```
{
  "data": "/9j/4AAQSkZJRgABAQAAQABAAD/2wBDAAIBAQEBAQIBAQECAgICAgQDAgICAgUEBAMEBgUGBgYFBgYGBwkIBgcJBWYGCAsICQoKCgoKBggLDAsKDAkKCgr/2wBDAQICAgUDAwUKBwYHCgoKCoKCgoKCgoKCgoKCgoKCgoKCgoKCgoKCgoKCgoKCgoK.....Urz7P/z71zeu/Zv0/cV0F5XN6xCf061EJczAis+v4Vt2c/7nNYFnBWjDBxXS3ymZuef7VRvJ6IO34VFL0/Co5Be93P/Z",
  "errCode": 0,
  "msg": "success",
  "successed": true
}
```

4.6.23 Front Camera Captured Depth Image in Base64 Format

Method: GET

URL: /real_time_data/ahl1d_rgb_img

Response Example:

```
{
  "data": "/9j/4AAQSkZJRgABAQAAQABAAD/2wBDAAIBAQEBAQIBAQECAgICAgQDAgICAgUEBAMEBgUGBgYFBgYGBwkIBgcJBWYGCAsICQoKCgoKBg.....PgpZ/wUR1j9sH9sn45/GT4C658QPB3w5+MniCyutX8Fax44vLv+2IbBI1s31CMzPHJslja4htSZIbHzRBbt5cKMfmCiiiiiiiv/2Q==",
  "errCode": 0,
  "msg": "success",
  "successed": true
}
```

4.6.24 Rear Camera Captured Depth Image in Base64 Format

Method: GET

URL: /real_time_data/ahl2d_rgb_img

Response Example:

```
{
  "data": "/9j/4AAQSkZJRgABAQAAQABAAD/2wBDAAIBAQEBAQIBAQECAgICAgQDAgICAgUEBAMEBgUGBgYFBgYGBwkIBgcJBWYGCAsICQoKCgoKBg.....PgpZ/wUR1j9sH9sn4
```

```
5/GT4C658QPB3w5+MniCyutX8Fax44vLv+2IbBI1s31CMzPHJs1ja4htSZIbHzRBbt5cKMf
mCiiiiiiiv/2Q==",
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

4.7 Movement Control

4.7.1 Set Movement Control Data

Method: POST

URL: /cmd/robot_move

Body:

```
{
  "speed": {
    "linearSpeed": 0.2, // x 轴线速度 m/s
    "slip_angle": 1.0, // 四转四驱底盘侧偏角 rad (范围 -2.14rad 到 2.14rad)
    "angularSpeed": 0.0 // 角速度 rad/s
  }
}
```

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "设置移动控制数据成功",
  "succeeded": true
}
```

Note:

- This interface needs to be called in a loop, with a request frequency greater than 10hz.

4.7.2 Set Movement Control Data - Stop

Method: POST

URL: /cmd/robot_move_stop

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "设置移动控制数据--关闭成功",
}
```

```
"succeeded": true
}
```

Note:

- This interface POST request has no body. If you want to stop the car immediately after calling “Set Movement Control Data”, you can call this interface, which sets the speed to 0. Or simply stop sending the “Set Movement Control Data” interface.

4.7.3 Fixed Distance and Fixed Angle Movement Control

Method: POST

URL: /cmd/move_with_params

Body:

```
{
  "linear_velocity": 0.2, // x 轴线速度 (m/s 范围-0.5 到 0.5)
  "slip_angle": 0.78, // 四转四驱底盘侧偏角 (rad 范围-2.14 到 2.14)
  "angular_velocity": 0.2, // 角速度 (rad/s 范围-0.5 到 0.5)
  "target_distance": 2, // 目标距离 (m)
  "target_angle": 3.14, // 目标角度 (rad 范围 0 到 3.14)
  "mode": 1 // 模式标志: 1=定距离移动, 2=定角度移动
}
```

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "Fixed distance movement completed",
  "succeeded": true
}
```

Note:

- Two modes: straight-line fixed distance movement and in-place fixed angle rotation. Both modes are available for differential drive and four-wheel-steering chassis; Ackermann chassis only has straight-line fixed distance movement. Four-wheel-steering chassis can use lateral mode in straight-line fixed distance movement.
- In straight-line fixed distance movement mode, the values of angular_velocity and target_angle will not take effect; in in-place fixed angle rotation mode, the values of linear_velocity, slip_angle, and target_distance will not take effect.
- For the four-wheel-steering chassis, when the sideslip angle slip_angle is not 0, lateral mode will be used to travel the target distance.

- When using this interface, ensure the car's localization is accurate.

4.7.4 Cancel Fixed Distance/Fixed Angle Movement Control

Method: POST

URL: /cmd/cancel_move_with_params

Body:

```
{  
  "cancel": true  
}
```

Response Example:

```
{  
  "data": null,  
  "errCode": 0,  
  "msg": "取消移动指令发布成功",  
  "succeeded": true  
}
```

Note:

- After calling the fixed distance/fixed angle movement control interface, if you do not want to continue moving, you need to call this interface to cancel.

4.8 Robot System Files

4.8.1 Download Robot Configuration File

Method: GET

URL: /file/robot_cfg_down

Note:

- Click “Send and Save Response” to save the configuration file.

4.8.2 Upload Robot Configuration File

Method: POST

URL: /file/robot_cfg_upload

Body:

body 中需要上传 file 配置文件

4.8.3 Update Robot Configuration File

Method: POST

URL: /cmd/robot_update_setting

Body:

```
{
  "avoid_obstacle": {
    "graph_follow": true, // 手绘路径是否避障。重启导航后生效，永久保存
    "record_follow": true, // 录制路径是否避障。重启导航后生效，永久保存
    "lidar_void_obstacle": true // 窄道区域是否启用雷达 scan true 启用, false 禁用
  },
  "max_speed": {
    "graph_follow": 0.2, // 手绘路径的最大速度，单位：m/s。最大 1.5m/s。立即生效，永久保存
    "record_follow": 0.5, // 录制路径的最大速度，单位：m/s。最大 1.5m/s。立即生效，永久保存
    "goal_follow": 0.5 // 自由导航的最大速度，单位：m/s。最大 1.5m/s。立即生效，永久保存
  }
}
```

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "修改 yaml 配置成功",
  "succeeded": true
}
```

4.8.4 Restore Robot Default Configuration

Method: POST

URL: /cmd/robot_reset_default

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "恢复默认配置成功",
  "succeeded": true
}
```

4.8.5 Download Map File

Method: GET

URL: /data/download_map?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	Yes	String	Map name

Response Example:

```
{
  "data": true,
  "errCode": 0,
  "msg": "下载成功",
  "succeeded": true
}
```

Note:

- Click “Send and Save Response” to download the map file compressed package to the local machine.
- The map file contains all point, path, and area data.

4.8.6 Upload Map File

Method: POST

URL: /data/upload_map?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	No	String	Map name

Body:

在 body 中上传地图文件压缩包，例如 map.zip

Response Example:

```
{
  "data": null,
  "errCode": -1,
  "msg": {},
  "succeeded": false
}
```

Response Example 2:

```
{
  "data": null,
  "errCode": -1,
  "msg": "地图名重复",
  "succeeded": false
}
```

Response Example 3:

```
{
  "data": null,
  "errCode": -1,
  "msg": "地图已存在!",
  "succeeded": false
}
```

4.8.7 APP Logo Image Download

Method: GET

URL: file/logo

4.8.8 APP Logo Image Upload

Method: POST

URL: /file/logo

Body:

在 body 中上传 png 图片

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "成功",
  "succeeded": true
}
```

4.8.9 Get APP Logo Base64 Data

Method: GET

URL: file/logo/base64

Response Example:

```
{
  "data": "/9j/4QCCRXhpZgAATU0AKgAAAAGABAEEAAMAAAABAAoAAAAEBAAMAAAABAc
UAAIdpAAQAAAABAAAAPgESAAMAAAABAAQAAAAAAAAAAAAAAAAZIIAAMAAAABAAAAAAAAAAAAAwEAA
AMAAAABAAoAAAAEBAAMAAAABAcUAAAEESAAMAAAABAAAAAAAAAAD/4AAQSkZJRgABAQAAAQAB
AAD/4gIoSUNDX1BST0ZJTEUAAQEAAAIIYAAAAAIIQAAbtbnRyUkdCIFhZWiAAAAAAAAAAAAAAAA
AAABhY3NwAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAQgiQhCCJCEIikIQgiQhCCJCE
IikIQgiQhCCJCEIikIQgiQhCCJCEIikIQgiQhCCJCEIikIQgiQhCCJCEIikIQgiQhCCJCEI
IkIQgiQhCCJCEIikIQgiQhCCJCEIikIQgi/9k=",
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

4.8.10 Download Map PNG Image

Method: GET

URL: /data/down_map_png?map_name={map_name}

Query:

Parameter	Required	Type	Description
map_name	No	String	Map name

Note:

- Click “Send and Save Response” to download the image.

4.8.11 Get APP Background Image Base64 Data

Method: GET

URL: file/background/base64

Response Example:

```
{
  "data": "/9j/4AAQSkZJRgABAQEAYABgAAD/2wBDAAwMCagMCAgMDAwMEAwMEBQgFBQ
QEBQoHBwYIDAoMDAsKCwsNDhIQDiiigAooooAKKKKACiigAooooAKKKKACmU+igBlFFFUA
UUUUAFFFFABRRRQAUUUUAFFFFABRRRQAUUUUAFFFFABRRRQAUFQFFFFABRRRQAU2nUUAOp1No
AKKKKACiigAooooA/9k=",
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

4.8.12 APP Background Image Upload

Method: POST

URL: /file/background

Body:

在 body 中上传图片

4.9 Robot Data

4.9.1 Restore Default Policy Configuration (Under development)

Method: POST

URL: /cmd/robot_reset_base_default

4.9.2 Get Navigation Configuration

Method: GET

URL: /data/robot_get_setting

4.9.3 Get Navigation System Status

Method: GET

URL: /data/sys_status

Response Example:

```
{
  "data": 3, // 导航状态
  "errCode": 0,
  "msg": "获取成功",
  "succeeded": true
}
```

Response Example 2:

```
{
  "data": 1, // 未运行
  "errCode": 0,
  "msg": "获取成功",
  "succeeded": true
}
```

Response Example 3:

```
{
  "data": 2, // 扫图
  "errCode": 0,
  "msg": "获取成功",
  "succeeded": true
}
```

Response Example 4:

```
{
  "data": 4, // 正在执行导航
  "errCode": 0,
  "msg": "获取成功",
  "succeeded": true
}
```

4.9.4 Get MAC Address

Method: GET

URL: /data/sys_mac

Response Example:

```
{
  "data": [
    "c4:83:72:10:cf:9c",
    "c4:83:72:10:cf:9d"
  ],
  "errCode": 0,
  "msg": "获取成功",
  "succeeded": true
}
```

4.9.5 Get APP Main Page Title Name

Method: GET

URL: /data/local_sys_info

Response Example:

```
{
  "data": {
    "sys_info": {
      "mian_page": {
        "title": "YUHESEN" // APP 主页标题名称
      },
      "test": 9999
    }
  },
  "errCode": 0,
  "msg": "获取成功",
  "succeeded": true
}
```

4.9.6 Modify APP Main Page Title Name

Method: POST

URL: /cmd/update_local_sys_info

Body:

```
{
  "mian_page": {
    "title": "YUHESEN"
  }
}
```

Response Example:

```
{
  "data": null,
  "errCode": 0,
  "msg": "修改本地配置成功",
  "succesed": true
}
```

4.9.7 Get About Us

Method: GET

URL: /data/robot_about_us

Response Example:

```
{
  "data": {
    "app": {
      "address": "深圳市龙岗区",
      "company": "深圳煜禾森科技有限公司",
      "soft_provider": "深圳煜禾森科技有限公司",
      "soft_tel": "0755-23356818",
      "tel": "0755-23356818",
      "version": "1.4.0" // 后端版本
    },
    "log_config": {
      "log_enc_key": "YHS@2023"
    },
    "map_config": {
      "map_free": 45,
      "map_occ": 55
    },
    "navigation": {
      "chassis_version": "FW-mid", // 底盘型号
    }
  }
}
```

```

        "version": "1.0.4" // 导航版本
    },
    "updater": {
        "base_dir": "/home/yhs/yhs-robot-updater",
        "package_dir": "/home/yhs/yhs-robot-updater/storage/current_update",
        "package_nav_dir": "/home/yhs/yhs-robot-updater/storage/current_update_nav",
        "port": 8889,
        "update_nav_route": "http://127.0.0.1:8889/updater/update_nav",
        "update_route": "http://127.0.0.1:8889/updater/update"
    }
},
"errCode": 0,
"msg": "获取成功",
"successed": true
}

```

4.9.8 Get Robot Appearance Dimensions

Method: GET

URL: /data/robot_show

Response Example:

```

{
    "data": "[[0.31,0.26],[0.31,-0.26],[-0.31,-0.26],[-0.31,0.26]]", // 机器人的四个顶点相对于旋转中心的偏移量 m
    "errCode": 0,
    "msg": "获取成功",
    "successed": true
}

```

4.9.9 Get Log List

Method: GET

URL: /log/get_all_logs?log_type={log_type}

Query:

Parameter	Required	Type	Description
log_type	Yes	String	Log type: operation operation log launch robot ros log

Parameter	Required	Type	Description
			drive_error driver error log rosbag robot data log

Response Example:

```
{
  "data": [
    {
      "created_at": "2023-07-13T09:59:50.277895",
      "filename": "operation2023-07-13.log",
      "md5": "8f9af383d42c8a7cc59c5e3447acf982",
      "modified_at": "2023-07-13T17:23:03.394586",
      "size": 559949
    },
    {
      "created_at": "2023-07-12T09:39:00.967297",
      "filename": "operation2023-07-12.log",
      "md5": "bb0fe8372570308ccccc765277e548b",
      "modified_at": "2023-07-12T18:00:08.630563",
      "size": 7984096
    },
    {
      "created_at": "2023-07-17T08:26:22.706345",
      "filename": "operation2023-07-17.log",
      "md5": "6851b40bf33e63d29ddbdfa7637eb89d",
      "modified_at": "2023-07-17T14:25:34.050967",
      "size": 2350954359
    }
  ],
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```

4.9.10 Download Log

Method: GET

URL: /log/down_logs?log_type={log_type}&file_name={file_name}

Query:

Parameter	Required	Type	Description
log_ty	Yes	Strin	Log type:

Parameter	Required	Type	Description
pe		g	operation operation log launch robot ros log drive_error driver error log rosvag robot data log
file_name	Yes	String	Log file name

4.9.11 Delete Log

Method: GET

URL: /log/drop_logs?log_type={log_type}&file_name={file_name}

Query:

Parameter	Required	Type	Description
log_type	Yes	String	Log type: operation operation log launch robot ros log drive_error driver error log rosvag robot data log
file_name	Yes	String	Log file name

Response Example:

```
{
  "data": "已成功删除日志文件!",
  "errCode": 0,
  "msg": "success",
  "succeeded": true
}
```